

8(a)	<i>any four from:</i> 1 Ca^{2+} act as second messenger ; 2 causes (K^+) channel proteins to open ; 3 X = potassium ions / K^+ (that leave the cell) ; 4 water potential of cell increases / water leaves cell by osmosis ; 5 (guard) cell, becomes flaccid / loses turgidity ;	4
8(b)	<i>any four from:</i> 1 (stomata) open to take up carbon dioxide for photosynthesis ; 2 (stomata) close to prevent, transpiration / (excessive) water loss ; 3 e.g. conditions when stomata close ; 4 (water homeostasis needed to) maintain turgidity of cells / supply water for reactions ; 5 <i>ref. to</i> daily rhythm or open during, day / light, and close, at night / at low light intensity / in the dark ; 6 AVP ;	4

Question	Answer	Marks
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